

Design Verification Plan and Report (virtual-test version)

All values mechanically taken from simulation outputs; uncovered conditions honestly marked N/A.

on item	Condition	Result	Determination	Source
arge capacity	1C CC discharge, 25 °C	6.372 Ah	—	`r5_d11_1c.json:capacity_al
c energy density	1C discharge energy ÷ layer-stack volume	1260.2 Wh/L	✓ ≥ 950	`r5_d11_energy.json:energy
lateau	discharge-time midpoint voltage	4.131 V	✓ ≥ 4.1	`r5_d11_energy.json:midpoi
harge temperature rise	4C charge, 45 °C ambient, lumped thermal	T_max 321.68 K	✓ ≤ 323.15 K	`r5_d11_4c.json:T_max_K`
harge plating	anode surface potential < 0 V	min +0.043 V → no plating	✓ false	`r5_d11_4c.json:anode_pot
hess after 100 cycles	100×1C cycles (isothermal)	193.6 nm	✓ ≤ 500 nm	`r5_d11_aging.json:sei_thic
tration	—	N/A (beyond pure-simulation boundary, requires physical experiment)	—	—
ge to thermal runaway	—	N/A (beyond pure-simulation boundary)	—	—
rop	—	N/A (requires physical experiment)	—	—
e life (>100 cycles)	—	N/A (not simulated)	—	—

Conclusion

- **Result: ALL simulated criteria PASS** (6/6 simulated items). Full pass per DFN-verified final candidate `LNMO-DFN-D11`.
- **Uncovered (limitations)**: nail penetration, overcharge-to-thermal-runaway, crush/drop, and cycle life beyond 100 cycles are outside the pure-simulation boundary and require physical experiments.
- **Honest note**: the 4C CC charge reaches ~27 % SOC before the 4.7 V cutoff (positive-electrode diffusion-limited, see design_spec.md §6 caveats).